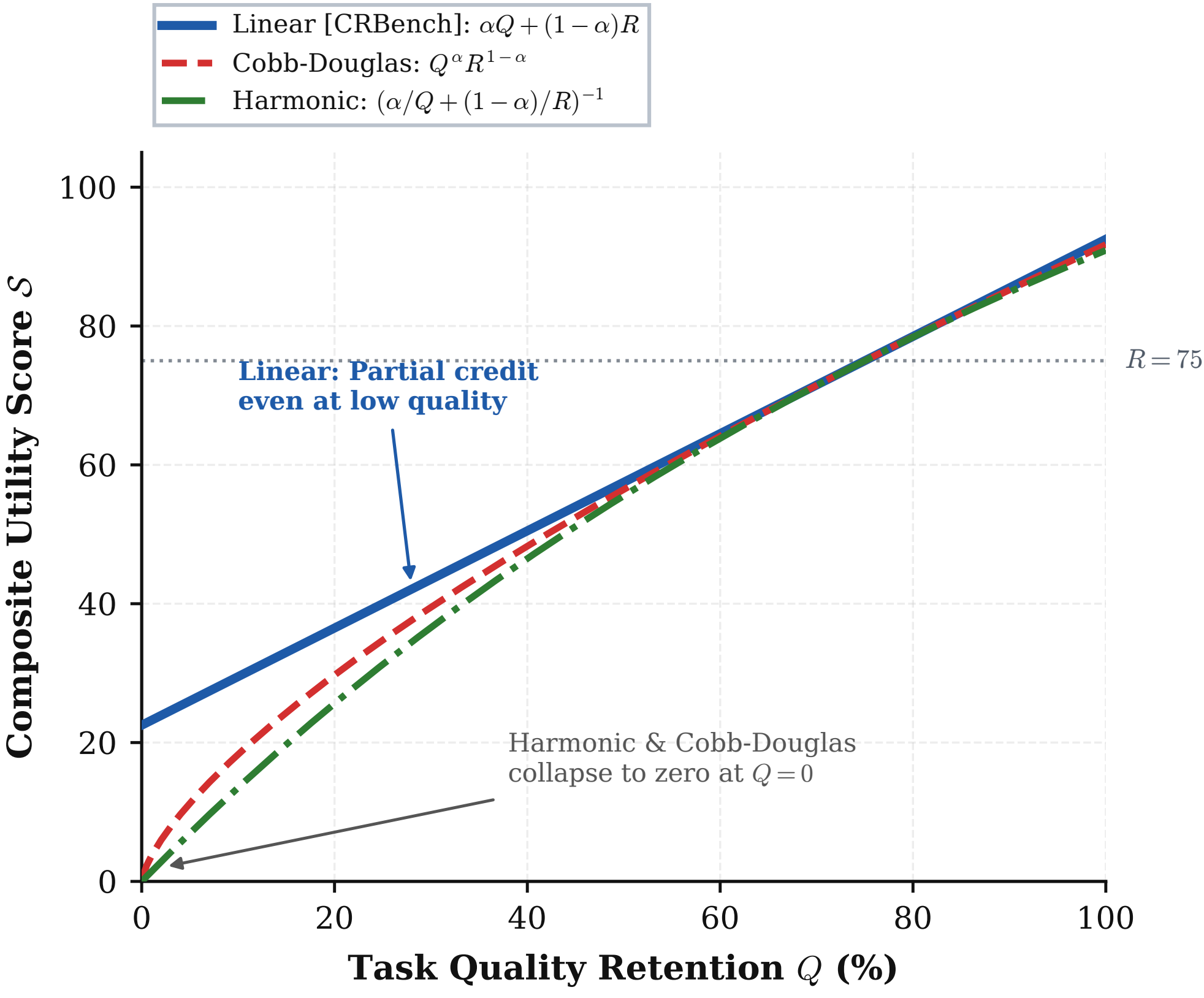
Linear [CRBench]: $\alpha Q + (1-\alpha)R$
Cobb-Douglas: $Q^{\alpha} R^{1-\alpha}$
Harmonic: $(\alpha/Q + (1-\alpha)/R)^{-1}$
Composite Utility Score $\mathcal{S}$
Task Quality Retention Q (%)
0
20
40
60
80
100
$R = 75$
Linear: Partial credit
even at low quality
Harmonic & Cobb-Douglas
collapse to zero at $Q = 0$